Telematics and Informatics Reports 20 (2025) 100271

Contents lists available at ScienceDirect

Telematics and Informatics Reports

journal homepage: www.elsevier.com/locate/teler

A machine learning framework to predict the severity and management of pediatric appendicitis using clinical markers and five explainable techniques

Pavanya Manohar [a], Anisha Palkar [b], Niranjana Sampathila [b], Venkatesh Bhandage [c], Krishnaraj Chadaga [c,*]

[a] Department of Obstetrics and Gynecology, Manipal Academy of Higher Education, Manipal, Karnataka, 576104, India
[b] Department of Biomedical Engineering, Manipal Institute of Technology, Manipal Academy of Higher Education, Manipal, Karnataka, 576104, India
[c] Manipal Institute of Technology, Manipal Academy of Higher Education, Manipal, India

ARTICLE INFO

Keywords:
Appendicitis
Explainable artificial intelligence
Management
Severity
Markers.

ABSTRACT

Appendicitis remains a common surgical emergency in children and adolescents, where timely diagnosis and accurate severity assessment are critical to prevent complications such as perforation and peritonitis. In this study, we applied machine learning (ML) approaches to clinical data to enhance prediction of both disease severity and the need for surgical versus conservative management. Compared with traditional logistic regression, ML models demonstrated improved predictive performance, particularly in identifying high-risk patients. The most influential variables included length of hospital stay, loss of appetite, C-reactive protein (CRP), presence of peritonitis, and the Alvarado score. These predictors, when analyzed through ML, allowed better stratification of patients into severe versus non-severe categories, achieving 94 % accuracy for severity prediction and 89 % accuracy for management prediction. Explainable AI techniques further clarified the role of each variable, ensuring interpretability for clinicians. Our findings highlight that ML-based decision support tools can complement conventional clinical assessment, potentially reduce diagnostic delays and supporting more precise treatment planning in hospital settings.

1. Introduction

Appendicitis is caused when the appendix swells or becomes infected [1]. Patients experience pain in the lower abdomen starting below the belly button. Most people develop appendicitis between the ages of 10–30 [2]. Appendicitis is treated by the surgical removal of the appendix or by administering antibiotics. This condition can be caused by various reasons, such as the formation of appendix stones, lymphoid hyperplasia, colitis, tumors, parasites, cystic fibrosis, appendiceal intraluminal tumors, foreign bodies and Enterobius vermicularis [3–5]. Identifying appendicitis symptoms can be crucial since many symptoms may go unnoticed. Symptoms include pain in the lower abdomen, nausea, fever, inability to pass gas, swollen belly, loss of appetite, severe cramps, and constipation [4]. If appendicitis is not treated, the bacteria can cause peritonitis, which can further lead to life-threatening sepsis [5]. The other complications of appendicitis are mentioned in Fig. 1. This research specifically focuses on children. At least 70,000 kids in the United States are affected by this condition every year [6]. Appendicitis can be diagnosed using imaging tests (Abdominal X-ray, ultrasonography (USG), computed tomography scans), blood and Urinalysis, among others. Several clinical scoring systems, such as the Alvarado score, Pediatric Appendicitis Score (PAS), and Appendicitis Inflammatory Response (AIR) score, are commonly used to assist in diagnosing appendicitis [7]. In some cases, they may help distinguish between perforated and simple appendicitis [7]. In most cases, surgeries are required. In some instances, the children might need antibiotics. The surgery can be either laparoscopy or laparotomy (open) [8]. Following surgery, children are required to remain in the hospital for at least a day [8]. However, in case of severe appendicitis (if the appendix has already perforated), the patient might be required to stay in the hospital for up to a week so that antibiotics and pain medication can be administered. Most children recover from the surgery quickly. Complications after surgery, such as infections, abscesses, and small bowel obstruction, have been observed in rare cases [9].

* Corresponding author.
E-mail address: Krishnaraj.chadaga@manipal.edu (K. Chadaga).

https://doi.org/10.1016/j.teler.2025.100271
Received 30 June 2025; Received in revised form 3 October 2025; Accepted 1 November 2025
Available online 3 November 2025
2772-5030/© 2025 The Author(s). Published by Elsevier B.V. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

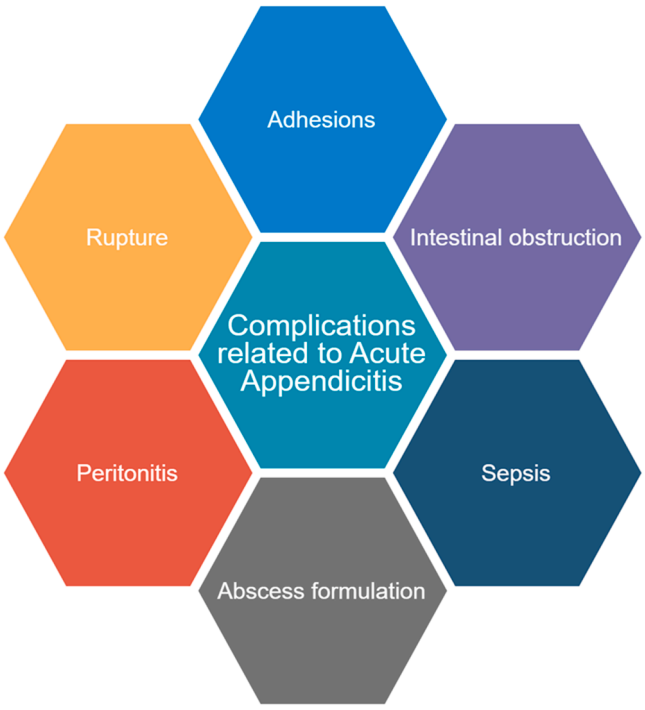


Fig. 1. Complications which occur if appendicitis is not treated.

Artificial intelligence (AI), Machine Learning (ML), and Explainable Artificial Intelligence (XAI) are thriving in the current decade. Their applications have been utilized by most domains, such as Engineering, Medicine, Science, Commerce, and others [10–12]. In medicine, the above technologies are constantly used in diagnosis, prognosis, resource allocation, forecasting, and other areas [13–15]. This study uses ML to predict the "Management" and "Severity" of pediatric appendicitis patients. Healthcare professionals are confident in using AI applications due to the birth of a new paradigm called Explainable Artificial Intelligence (XAI). XAI makes the blackbox algorithms more explainable by analyzing the contribution made by each feature [16,17]. A commoner can understand the predictions since it makes use of visual descriptions and simple graphs.

Several studies have used AI applications for appendicitis. Recent studies are reviewed below. Marcinkevics et al. [18] designed an explainable model to diagnose pediatric appendicitis using ultrasonography. This research considered one thousand seven hundred nine ultrasound images of 579 patients. The customized model acquired a highest AUC of 0.80. In another study, AI was used to grade pediatric appendicitis using ultrasound reports of 2100 children [19]. A maximum precision of 97.09 % was obtained in this study. Byun et al. [20] used an algorithm to forecast complex case of appendicitis by analyzing CT, clinical, and laboratory markers.

A decision tree (DT) model was developed for the dataset containing the details of 315 children. An AUC of 0.91 was obtained in this clinical research. The most critical markers were C-reactive protein (CRP), white blood cells (WBC) and temperature. Another study made use of a clinical decision support system to predict appendicitis early [21]. One hundred eighty-one patients were considered in this research, and the decision support framework made use of a hybrid framework. The combined classifier obtained an accuracy of 95 %. Males et al. [22] developed an explainable machine learning model based on data from 551 pediatric patients. The model significantly reduced negative appendectomies by approximately 17 % and showed high sensitivity (0.997), outperforming traditional scoring systems like the AIR score across most sensitivity-specificity trade-offs.

From the above studies, it can be seen that AI has been successfully used in the battle against pediatric appendicitis. However, in most studies, the decisions made by the models were difficult to decipher and understand. The aim of the current research is to predict the management and severity of pediatric appendicitis and to explain the results to the stakeholders. This study strives to create a more transparent, explainable, interpretable, and trustworthy model. The contributions made by our research on the existing literature are listed below:

- Pearson's correlation and mutual information algorithms have been used to predict the important features that can aid in appendicitis management and severity.
- A customized stacking algorithm is designed using multiple bagging, boosting, and other baseline algorithms. Each model was further optimized using the grid search hyperparameter tuning technique.
- Five XAI techniques namely (a) Shapley Additive Values (SHAP), (b) Local Interpretable Model-Agnostic Explanations (LIME), (c) Eli5, (d) QLattice, and (e) Anchor are used to demystify the predictions.
- An extensive discussion about the essential markers from a clinical view-point.

2. Materials and methods

2.1. Data description

The data used in this pediatric medical research was gathered from the "Children's Hospital St. Hedwig" in Regensburg, Germany and is available online [23]. The patient details include physical examination

Table 1
Description of the pediatric appendicitis dataset.

Patient Attribute	Description of the Attribute	Patient Attribute	Description of the Attribute	Patient Attribute	Description of the Attribute
1. Age	In years	21. White blood cell (WBC) count	Leucocyte count.	41. Pathological lymph nodes	Whether lymph nodes have become enlarged?
2. Sex	Male or Female	22. Red blood cell (RBC) count	Erythrocyte count.	42. Lymph node location	Location of the lymph node.
3. Height	In centimeter	23. Hemoglobin	Hemoglobin percentage in blood.	43. Bowel wall thickening	Edema in the intestinal wall
4. Weight	In kilogram	24. RDW (Red Cell Distribution Width)	Size and volume of RBC.	44. Ileus	Temporary absence of peristalsis
5. Body mass index	Anthropometric measurement.	25. Thrombocyte count	Platelets present in the blood.	45. Coprostasis	Fecal impaction in the intestine i.e. colon
6. Length of stay	Total number of days in hospital	26. Neutrophil percentage	It is a type of WBC, expressed in percentage	46. Meteorism	Excess gas in the intestine
7. Alvarado score	Scoring method that diagnose acute appendicitis in adult.	27. Neutrophilia	Increased number of neutrophils	47. Enteritis	Whether enteritis exists?
8.Pediatric appendicitis score	Clinical scoring system in children for appendicitis	28. Segmented neutrophils	Matured neutrophils count.	48. Apendicolith	Fecal deposits in the appendix
9. Peritonitis	Inflammation in the peritoneal cavity	29. C-reactive protein	It increases during inflammation or infection	49. Perforation	USG suggestive sign in appendix
10. Migratory Pain	Localization of pain in the abdomen	30. Ketones in Urine	Ketones present in urine.	50. Appendicular abscess	Whether Appendiceal mass exists?
11. Lower right abdominal pain	Presence or absence of pain in the lower right abdomen.	31. RBC in urine	Appearance of RBC in urine	51. Abscess location	Location where abscess exists.
12. Contralateral rebound tenderness	Presence or absence of pain in the opposite side of the abdomen	32. WBC in urine	Appearance of WBC in urine	52. Conglomerate of bowel loops	Sign of intraperitoneal inflammation
13. Ipsilateral rebound tenderness	When pressure is released from lower abdomen, pain appears on the same side.	33.Abdominal USG	Whether abdominal USG is performed?	53. Gynecological findings	Gynecolological abnormalities.
14.Pain during coughing	Abdominal pain during coughing.	34. Appendix on USG	Diagnosis of veriform appendix in USG	54. Ultrasound images	Ultrasound images of the appendix.
15. Psoas sign	Abdominal pain during hip extension.	35. Appendix diameter	Diameter of the appendix.	55. Diagnosis presumptive	Preliminary/ Initial diagnosis.
16. Nausea	An urge to vomit	36. Free fluids	Presence or absence of free fluids in the abdomen	56. Diagnosis	Actual diagnosis
17. Loss of appetite	Loss of hunger	37. Appendix wall layers	Characteristics of appendix layer	57. Management	Managing appendicitis using various treatments.
18. Body temperature	Measured in Celsius	38. Target sign	USG sign in appendix	58. Severity	Severity level of appendicitis.
19. Dysuria	Pain during urination	39. Perfusion	Blood flow to the appendix wall		
20. Stool	Bowel movement	40. Surrounding tissue reaction	Inflammation surrounding appendix		

results, biomarkers, and specific laboratory data. It consists of 58 features and diagnostic outcomes for 782 patients. Every marker and its corresponding values are clearly described in the metadata (embedded in the Excel file) [23]. This dataset contained the following four classifications: "presumptive diagnosis," "diagnosis," "management", and "severity." This research focuses only on "management" and "severity". "Management" consisted of four categorical values: "Conservative" (encoded as one) (management with antibiotics), primary surgical (encoded as 2) (surgery on admission), surgical (encoded as 3) (surgery after prior stay) and simultaneous appendicitis (encoded as 4) (appendicitis with another abdominal surgery). The label "severity" consisted of two categorical values: "Complicated (encoded as 0)" (Perforated appendix) and "Uncomplicated" (encoded as 1) (Non perforated appendix). The gathered dataset contained a large number of null values. Additionally, there was a slight imbalance in the dataset. There were 120 complicated cases and 662 uncomplicated cases. In the case of managing appendicitis, a total of 480 patients received conservative treatment. Additionally, 270 patients underwent surgery upon admission, while 27 patients required surgery after a prior hospital stay. There was just one patient who had appendicitis with another abdominal surgery. In this study, we removed the category "simultaneous appendicitis" since there was only one patient in that category. Machine learning results are not reliable when the dataset is too small. Overfitting and biased results can also be an issue. Further, unbalanced classes lead to erroneous results. Table 1 describes the attributes in the study.

2.2. Data preprocessing

Initially, target variables “Diagnosis” and “Presumptive diagnosis” were removed since our study focuses on “management” and “severity”. The median of the relevant marker was used to replace the null values that were present in the dataset. This procedure is crucial since a few models fail to run if null values exist [24]. The data was scaled in the range of zero to one using the normalization technique. Scaling is important to maintain data uniformity since the models favor the markers with higher values [25]. To eliminate model bias and ambiguity, we utilized one-hot encoding to represent categorical variables [26]. The patient data was also distributed into 80:20 ratio for training/testing purpose. Data balancing is compulsory in ML since the models supports the class with higher events. Borderline SMOTE was employed in the current work to balance the appendicitis and non-appendicitis classes [27]. To protect the integrity of the prediction, the testing dataset was not handled by Borderline SMOTE. The classifiers were ensembled using the stacking technique, which enhances the predictions by ensembling the classifiers using a meta classifier. The stacking methodology is depicted in Fig. 2. Each classifier was further enhanced using the grid search technique [28]. The best hyperparameters were chosen after iterating through the model’s search space.

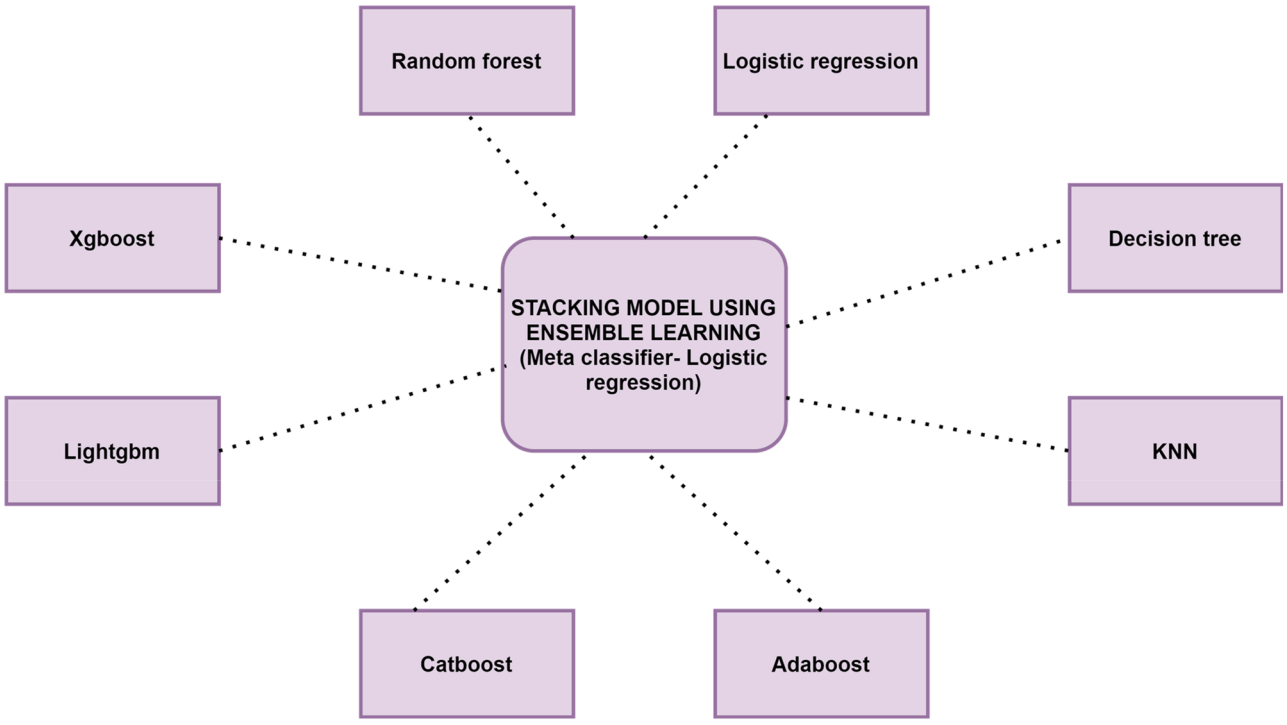


Fig. 2. Stacking ensemble algorithm for appendicitis management and severity.

2.3. Identifying important markers using feature importance

Mutual information and Pearson's correlation have been used to understand the important attributes for both "Management" and "Severity" use cases. Mutual Information algorithm utilizes the theory of information gain and entropy [29]. When the entropy of a marker is less, its information gain increases, making it a crucial attribute. Information gain is a value that ranges from 0–1. Values closer to 1 indicate that the markers are more important. Fig. 3 indicates the important markers chosen by mutual information in a bar graph format. The markers are represented in descending order based on their usefulness on the x-axis. The information gain is represented on the y-axis. For appendicitis management, the crucial markers are "Length of stay," "body temperature," "C-reactive protein (CRP)," "Height," and "Red blood cell distribution width (RDW), with "Length of stay" being the most important. For appendicitis severity, the critical markers include "Length of stay," "Body temperature," "Alvarado score," "Pediatric appendicitis score," and "CRP." In both use cases, the length of a patient stay is critical for the model outcome.

Pearson's correlation is used to identify the correlation between two variables [30]. The Pearson's correlation coefficient ranges from '−1′ to '1′. If the value is close to 1, it means the variables are positively correlated, and if the value is close to −1, the variables are negatively correlated. The variables are not related if the value is near 0. The Pearson's correlation heat map for management and severity are depicted in Fig. 4 and Fig. 5, respectively. For appendicitis management, attributes such as "Length of stay,"" peritonitis," "appendix diameter," "Alvarado score," and "CRP" are crucial. For appendicitis severity, attributes such as "Length of stay," "CRP," "WBC count," "Peritonitis," and "Alvarado score."

3. Results

Supervised learning algorithms and a customized stacking ensemble technique were tested for prediction [31,32]. The model predictions were further demystified using five heterogenous XAI techniques [33]. The ML pipeline used in this research is illustrated in Fig. 6.

3.1. Severity

The results obtained by the ML algorithms for severity prediction are tabulated in Table 2. Maximum accuracy of 94 %, was achieved by random forest, adaboost, catboost and xgboost. The ensemble stacking algorithm was able to get an accuracy of 93 %. Further, the optimal hyperparameters chosen by the grid search method are made in Table 3. The AUC curve, precision-recall curve, and confusion matrix are depicted in Fig. 7.

For its high reliability, the ensemble stacking model was utilized in XAI analysis. SHAP employs game theory to explain the predictions [34]. The models are interpreted by analyzing the contribution of each feature to the model output. The beeswarm plot depicted in Fig. 8 explains the variations in the markers for complicated and uncomplicated patients. Complicated cases are grouped toward the left of the hyperplane, and uncomplicated cases are grouped toward the right.

The markers are arranged in descending order of importance, with the best feature at the top. The most important markers are length of stay, loss of appetite, CRP, peritonitis, body temperature, Alvarado score, and pediatric appendicitis score. The length of stay, CRP, body temperature, Alvarado score, and pediatric appendicitis score are higher in the complicated appendicitis cohort. Peritonitis value is higher in the uncomplicated appendicitis cohort. Local interpretations (individual patient prediction explanation) can be made using force plots, as described in Fig. 9. Markers such as CRP and length of stay push the predictions towards complicated appendicitis.

LIME uses instance selection, data perturbation, weight assignment, and linear regression to train an interpretable model [35]. LIME explanations for complicated and uncomplicated appendicitis patients are depicted in Figs. 10(a) and 10(b), respectively. The figures represents the most significant features such as loss of appetite, length of stay, peritonitis, nausea, pediatric appendicitis score, and body temperature.

Eli5 is a popular Python library that interprets predictions [36]. It considers feature importance, weights, and bias to predict essential features. The Eli5 explanations are depicted in Fig. 11. Length of stay, age, CRP, and red blood cell count are crucial to differentiate between the two classes, according to Eli5.

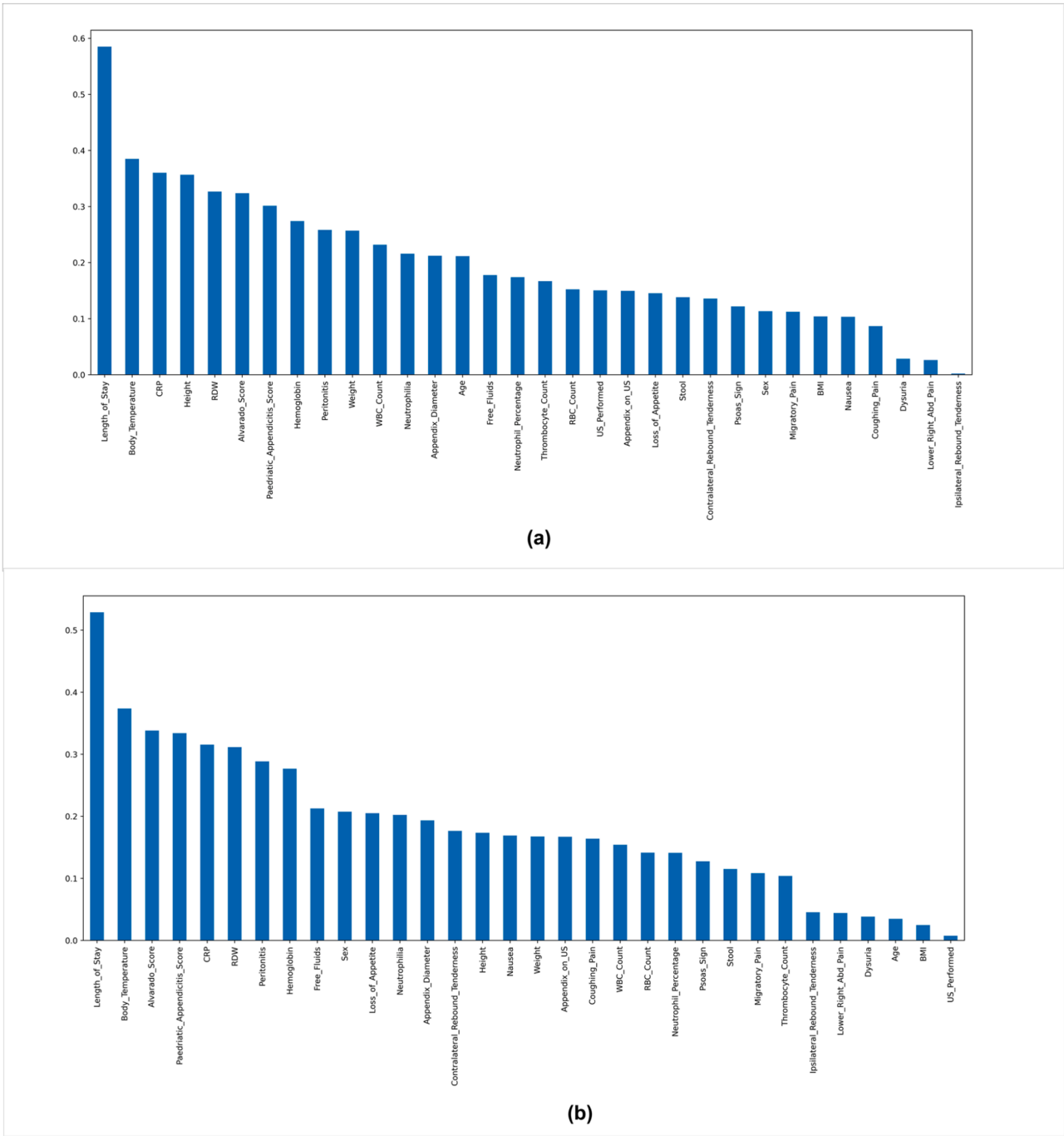


Fig. 3. Mutual information to identify the best features (a) Management (b) Severity.

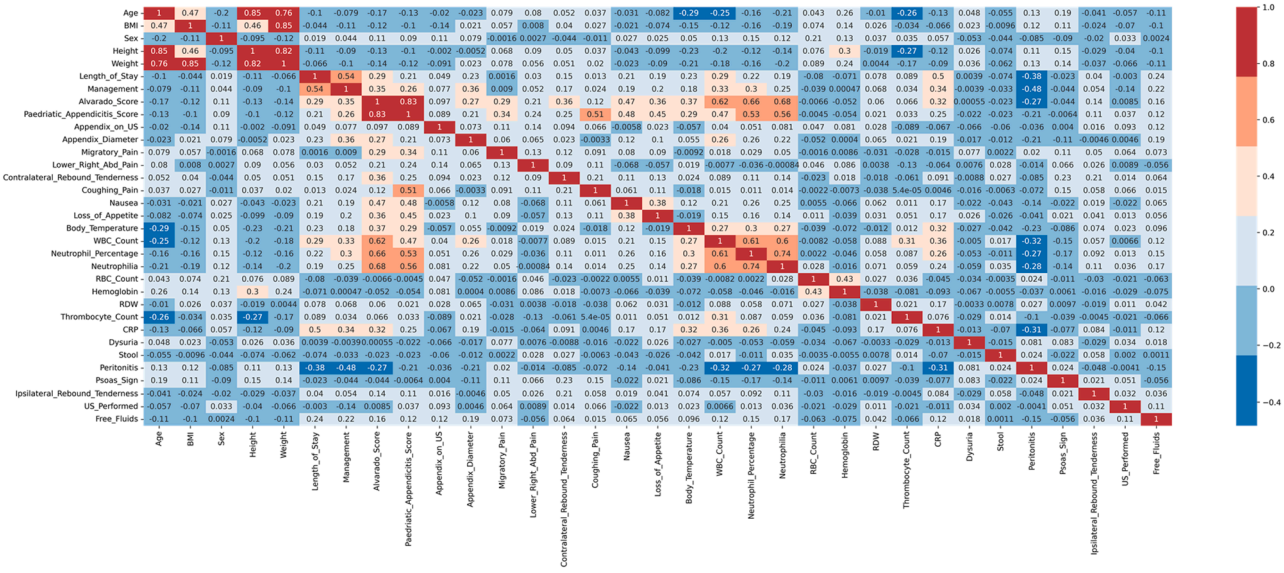


Fig. 4. Pearson's correlation heatmap to identify the best features for management use case.

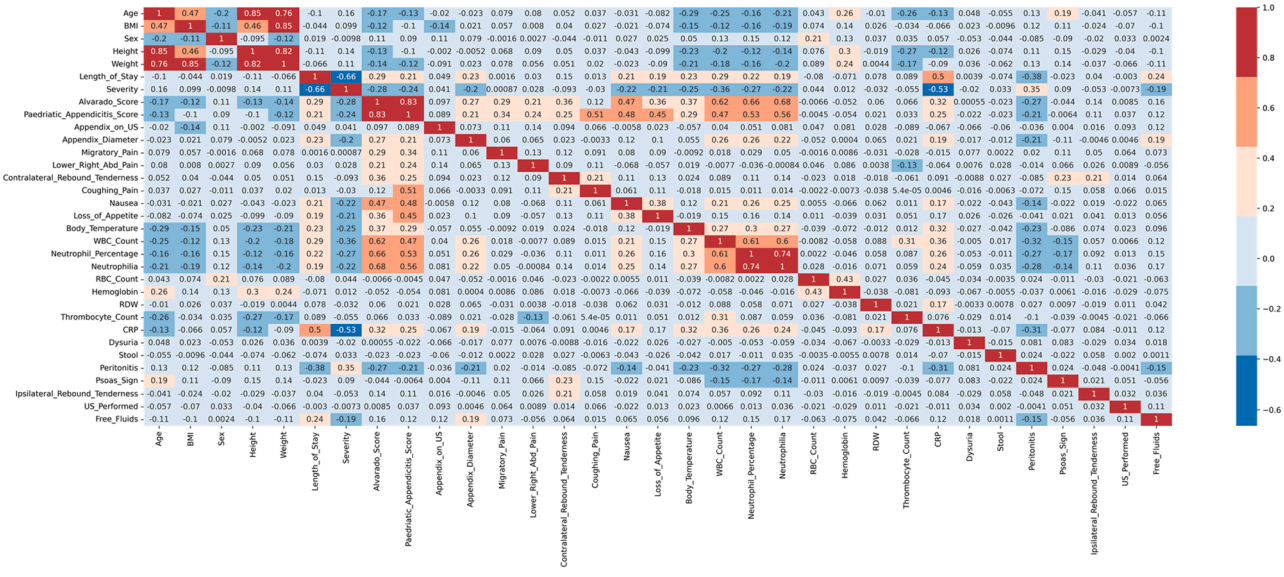


Fig. 5. Pearson's correlation heatmap to identify the best features for severity use case.

QLattice uses quantum computing to make the models more transparent [37]. Every attribute is defined as a register in the QLattice methodology. The data is trained using activation functions such as addition, multiplication, and other trigonometric functions. A sample Qgraph is depicted in Fig. 12. According to it, essential features include length of stay and white blood cell count.

The final XAI method employed in this investigation was anchor. It generates conditions that can be used to understand important features [38]. Further, the reliability of the condition is checked using two measures: Precision and coverage. Precision measures the accuracy of the condition, and coverage represents the number of instances that use the same condition. The anchor explanations for complicated and uncomplicated appendicitis are tabulated in Table 4. According to Anchor, the crucial markers are CRP, loss of appetite, length of stay, and nausea.

Five XAI techniques were used, and according to them, the most important markers for appendicitis severity prediction are length of stay, loss of appetite, CRP, peritonitis, body temperature, Alvarado score, pediatric appendicitis score, nausea, body temperature, RBC and WBC.

3.2. Management

The classification results for all the models are tabulated in Table 5. RF and the STACK model obtained an accuracy of 89 %. The catboost and RF models obtained a maximum accuracy F1-score of 89 %. Fig. 13 depicts the confusion matrix obtained by the ensemble stack. From the figure, it can be seen that most predictions are accurate.

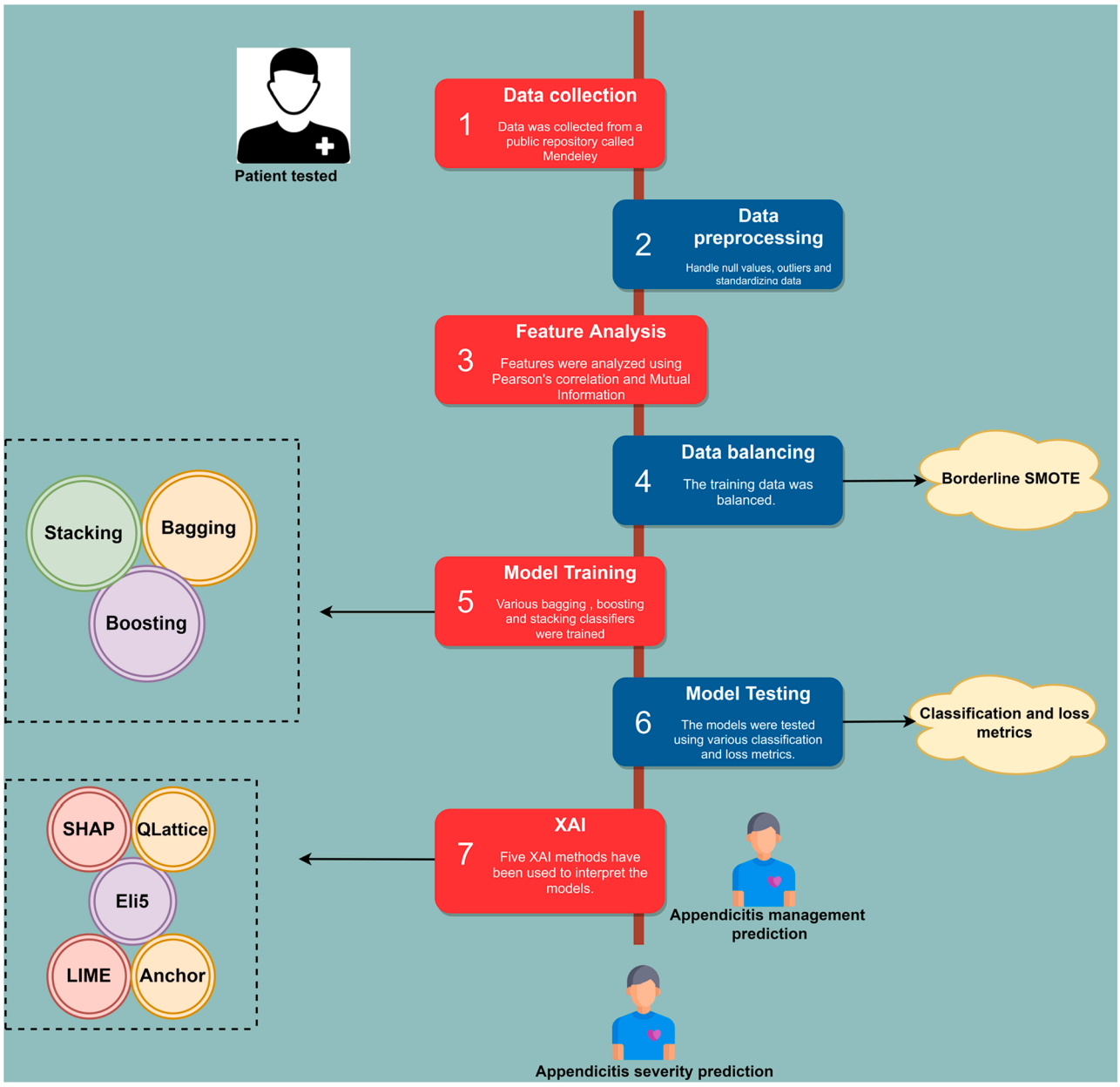


Fig. 6. Machine learning pipeline for appendicitis severity prediction and management.

Table 2
Appendicitis severity classification results.

Algorithm	Accuracy	Precision	Recall	F1-score	AUC	Hamming loss	Jaccard score	Log loss	MCC
RF	94	89	88	88	0.96	0.05	0.93	1.97	0.76
LR	92	83	90	86	0.96	0.07	0.91	2.63	0.73
DT	87	76	89	80	0.92	0.12	0.85	4.39	0.63
KNN	80	64	68	65	0.68	0.20	0.77	7.03	6.31
Adaboost	94	86	89	88	0.96	0.06	0.92	2.19	0.75
Catboost	94	87	91	89	0.97	0.05	0.93	1.97	0.78
Lightgbm	92	84	87	85	0.96	0.07	0.91	2.63	0.70
Xgboost	94	88	85	87	0.96	0.06	0.92	2.19	0.73
Ensemble stack	93	87	83	85	0.96	0.07	0.92	2.41	0.70

Table 3
Hyperparameters chosen by the algorithms to predict appendicitis severity.

Algorithm	Hyperparameters
RF	{'bootstrap': True, 'max_depth': 100, 'max_features': 4, 'min_samples_leaf': 3, 'min_samples_split': 9, 'n_estimators': 200}
LR	{'C': 100, 'penalty': 'l2'}
DT	{'criterion': 'entropy', 'max_depth': 15, 'max_features': 'auto', 'min_samples_leaf': 5, 'min_samples_split': 50, 'splitter': 'best'}
KNN	{'n_neighbors': 1}
Adaboost	{'learning_rate': 0.1, 'n_estimators': 200}
Catboost	{'border_count': 32, 'depth': 3, 'iterations': 250, 'l2_leaf_reg': 5, 'learning_rate': 0.03}
Lightgbm	{'lambda_l1': 0, 'lambda_l2': 1, 'min_data_in_leaf': 30, 'num_leaves': 31, 'reg_alpha': 0.1}
Xgboost	{'colsample_bytree': 0.4, 'gamma': 0.1, 'learning_rate': 0.05, 'max_depth': 3, 'min_child_weight': 1}
STACK	use_probas=True, average_probas=False, meta_classifier=Logistic regression

The ensemble stacking model was employed for XAI analysis due to its generally reliable results. SHAP was the first explainer used, and the bar plot obtained by it is depicted in Fig. 14. The attributes are organized in order of their contribution, starting from the highest to the lowest. According to it, the most important markers for appendicitis management prediction are the length of stay, peritonitis, loss of appetite, appendix diameter, and age. LIME was the following explainer used, and the prediction for a conservative management patient is explained in Fig. 15. Markers such as length of stay, peritonitis, diameter of appendix, thrombocyte count, and neutrophil percentage have been crucial. Fig. 16 explains the predictions of Eli5, the final explanation that was used.The most vital markers are the length of stay, peritonitis, thrombocyte count, and age. Qlattice and anchor were not used to explain appendicitis management since they did not support multi-classification analysis when writing this manuscript. According to three XAI techniques, the crucial features that can help in appendicitis management prediction are length of stay, peritonitis, loss of appetite, appendix diameter, age, thrombocyte count, and neutrophil percentage.

The essential markers chosen by feature analysis techniques, such as mutual information and Pearson's correlation, are compared with XAI techniques such as SHAP, LIME, Eli5, QLattice, and Anchor. The features they chose are tabulated in Table 6. It can be seen that the features selected by both the feature analysis and XAI techniques are almost similar. This further makes the predictions more robust and reliable.

4. Discussion

This study used ML and XAI to predict the severity and management of pediatric appendicitis patients. The dataset was also subjected to feature analysis before model deployment. Random forest, Adaboost, Catboost, and Xgboost obtained 94 % accuracy while classifying appendicitis severity. An 89 % accuracy was obtained by the random forest and the ensemble stack to predict pediatric appendicitis management. Further, essential marker variations were analyzed using five XAI techniques. The models can be used in real-time in hospital settings to aid doctors and other healthcare professionals.

When predicting appendicitis severity, it was observed that the length of stay was higher in complicated appendicitis patients. This agrees with already published studies since patients must be administered antibiotics if the appendix has already perforated [39–41]. The patient must be in observation and requires adequate bed rest. Loss of appetite was also observed in severe appendicitis patients. CRP levels were also higher in the severe cohort. Peritonitis was observed in both cohorts. The body temperature, Alvarado score, and pediatric appendicitis score were higher in the severe cohort. Alvarado score is a metric to measure the severity of appendicitis in both adults and children. Pediatric appendicitis score is a measure that indicates the severity of appendicitis exclusively in children. White blood cell count, neutrophil percentage, and appendix diameter were higher in the severe cohort. Nausea was also observed in complicated appendicitis patients. Other appendicitis studies have also observed all the above marker variations [39,40]. When predicting appendicitis management, the most critical markers were the length of stay, peritonitis, loss of appetite, appendix diameter, age, thrombocyte count, and neutrophil percentage. This study agrees with already published appendicitis research [40,41].

A few studies have used clinical markers and ML to predict appendicitis management/severity. Byun et al. [20] used ML to predict complicated appendicitis in pediatric patients using laboratory, CT, and clinical attributes. An AUC and sensitivity of 0.91 and 91.8 % were obtained. According to feature importance, CRP, white blood cells, and body temperature were important. Similar results have also been observed in this research. In another study, complications after pediatric appendectomy were predicted using wearable data [22]. The random forest obtained an accuracy of 83 %. Marcinkevics et al. [42] used ML for diagnosis/management/severity of pediatric appendicitis patients. A precision-recall curve of 0.92 and 0.70 were obtained for management and severity. Random forest was utilized for feature importance, and

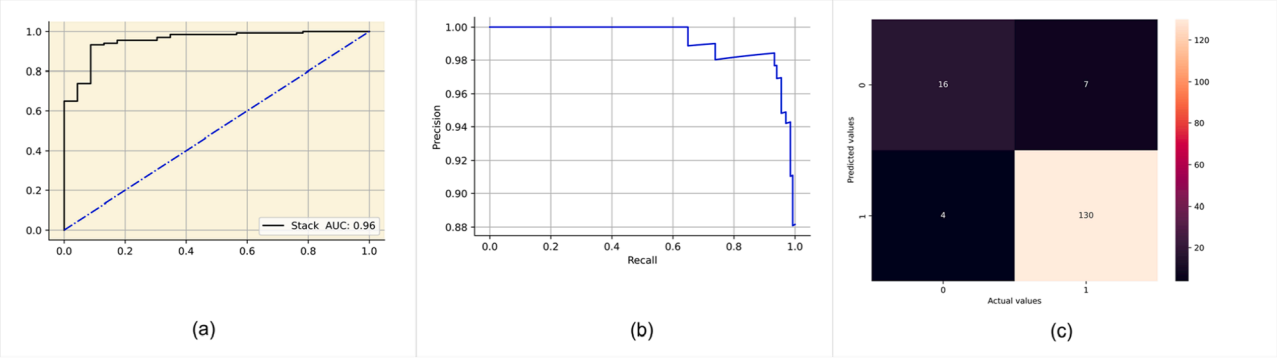


Fig. 7. (a) AUC; (b) Precision-recall curve; (c) Confusion matrix – obtained by the optimized STACK model.

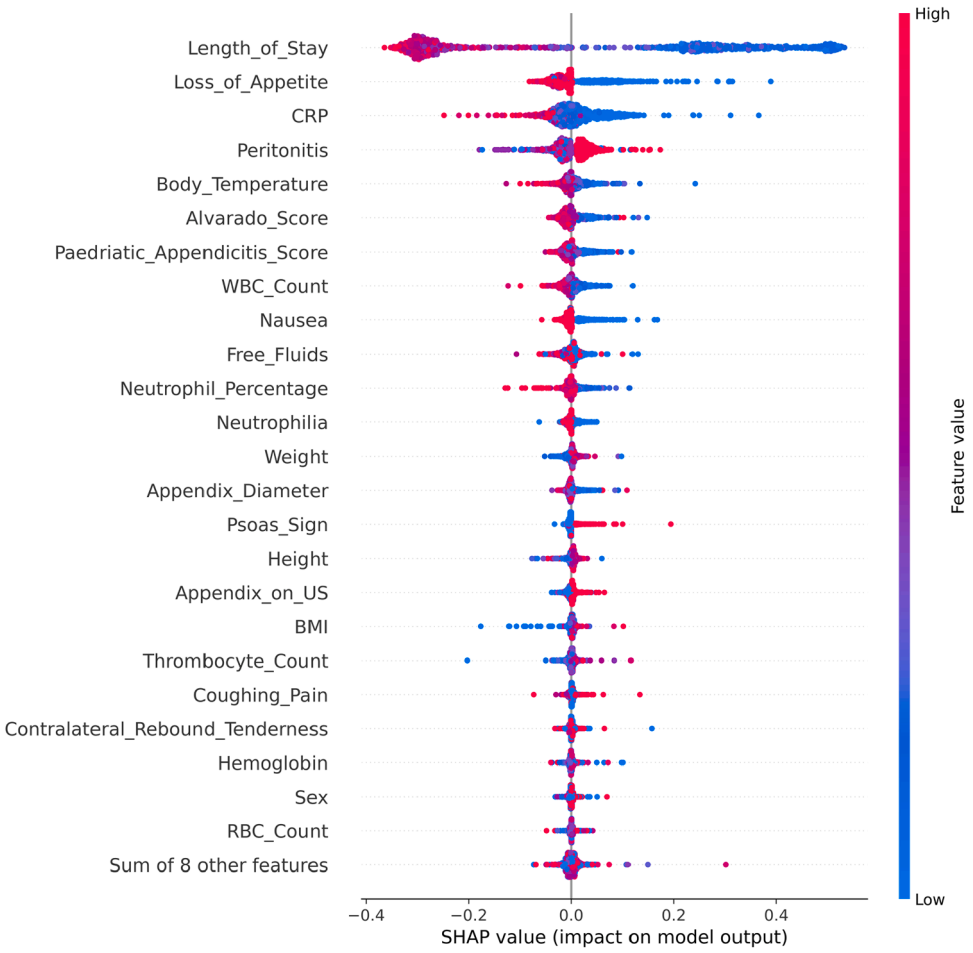


Fig. 8. SHAP beeswarm plot.



Fig. 9. SHAP force plot.

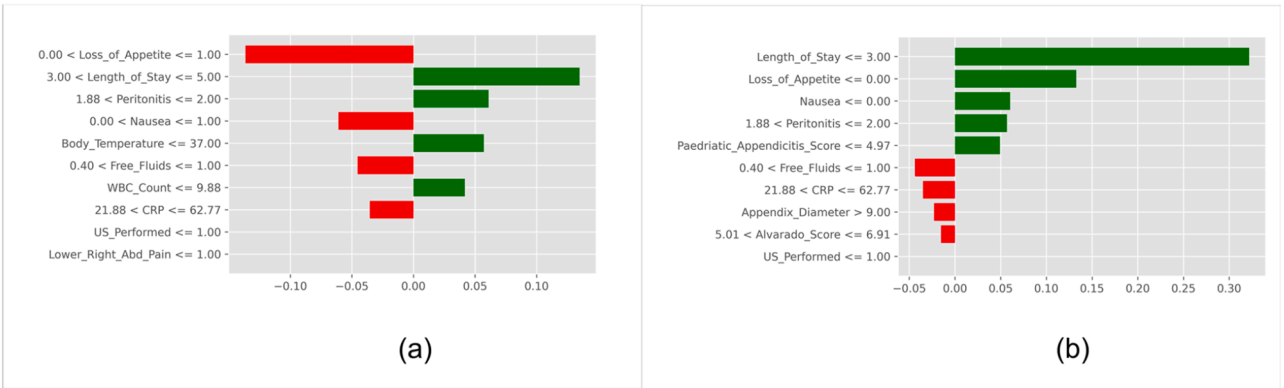


Fig. 10. LIME explanations (a) severe appendicitis patient (b) non-severe appendicitis patient.

y=1.0 (probability **1.000**) top features

Contribution[?]	Feature	Value
+0.500	<BIAS>	1.000
+0.361	Length_of_Stay	5.000
+0.094	Age	13.830
+0.027	CRP	1.000
+0.018	RBC_Count	4.160
+0.018	BMI	17.300
+0.017	Thrombocyte_Count	336.000
-0.035	Hemoglobin	12.600

Fig. 11. Eli5 explanations.

according to it, the essential markers were appendix diameter, peritonitis, WBC count, neutrophil percentage, and CRP. The following observations have also been recorded in this study. A detailed comparison of other studies with our research is made in Table 8. Most studies have not used multiple XAI techniques to predict the management/severity of appendicitis at the time of writing this manuscript. A few studies have used feature importance techniques to interpret the models.

This study had a few limitations. The number of patients was 782. The ML algorithms perform well when the dataset size increases. Deep learning models were not utilized since the dataset was small. In this study, medical validation was not performed, and the models were not deployed in real-time since this was a retrospective study. Feature selection was not performed, and all the features were included in this study. Some variables are related to each other and there might be some bias in the explanation of the results. The models were only trained and tested on a single dataset (n = 782). External validation on a different dataset is crucial to assess generalizability. Statistical analysis has also not been conducted for the publicly available dataset. In the near future, new datasets with a larger cohort could be collected and external validation could be performed. Federated learning, unsupervised learning, and reinforcement techniques could be included and compared. Datasets from multiple sources could be included for better reliability. The models could also use a cloud infrastructure. Feature importance techniques could be compared with XAI techniques. New deep-learning architectures could be trained and tested. Patient data could be protected using cryptographic and other information security measures. The models could be validated by the doctors and deployed in real-time to assist healthcare personnel. A simple user interface could be built to make predictions easily. Using 58 features without feature selection (other than mutual information and correlation analysis) increases the risk of multicollinearity. A more rigorous feature reduction method (e.g., PCA or recursive feature elimination) should be considered.

5. Conclusion

In this interdisciplinary research, artificial intelligence techniques were used to predict and manage pediatric appendicitis severity. The dataset consisted of 782 patients from Children's Hospital St. Hedwig" in Regensburg, Germany. Prior to training the classifiers, mutual information and Pearson's correlation techniques were employed for feature analysis. Multiple supervised learning techniques, including an ensemble stack, were used to predict the two use cases. A maximum accuracy of 94 % was obtained for appendicitis severity prediction, and a maximum accuracy of 89 % was obtained for appendicitis management prediction. According to the explainers, important markers include length of stay, loss of appetite, CRP, peritonitis, Alvarado score, and pediatric appendicitis score. The markers obtained in this research also aligned with existing appendicitis medical studies. Hence, the ML models could be used to assist healthcare personnel in the battle against pediatric appendicitis.

Table 4
Anchor explanations for both complicate and uncomplicated appendicitis patients.

Patient type	Anchor condition	Precision	Coverage
Complicated	CRP ≤ 4.00 AND Loss_of_Appetite ≤ 0.00	0.94	0.13
Complicated	Length_of_Stay > 5.00 AND CRP > 21.88	0.84	0.36
Complicated	Length_of_Stay > 7.84 AND Peritonitis ≤ 1.88	0.84	0.18
Complicated	Length_of_Stay ≤ 5.00 AND CRP ≤ 4.00	0.99	0.23
Complicated	Length_of_Stay ≤ 5.00 AND CRP ≤ 4.00	0.98	0.22
Uncomplicated	Length_of_Stay ≤ 7.84 AND CRP ≤ 4.00	0.96	0.24
Uncomplicated	Length_of_Stay ≤ 3.00 AND Free_Fluids ≤ 0.00	0.96	0.22
Uncomplicated	Length_of_Stay ≤ 3.00 AND CRP ≤ 4.00	1	0.17
Uncomplicated	Length_of_Stay ≤ 3.00 AND Nausea ≤ 0.00	1	0.16
Uncomplicated	Length_of_Stay ≤ 3.00 AND Nausea ≤ 0.00	0.99	0.17

Table 5
Appendicitis management classification results.

Algorithm	Accuracy	Precision	Recall	F1-score	Hamming loss	MCC
RF	89	87	89	88	0.10	0.76
LR	73	81	73	77	0.26	0.50
DT	67	73	67	70	0.33	0.38
KNN	61	65	61	63	0.38	0.25
Adaboost	83	83	83	83	0.16	0.65
Catboost	88	88	88	88	0.12	0.75
Lightgbm	87	85	87	86	0.12	0.72
Xgboost	86	85	85	85	0.14	0.72
STACK	89	86	89	87	0.11	0.75

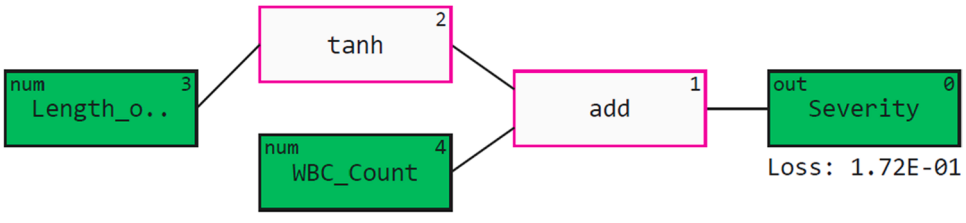


Fig. 12. QLattice interpretations.

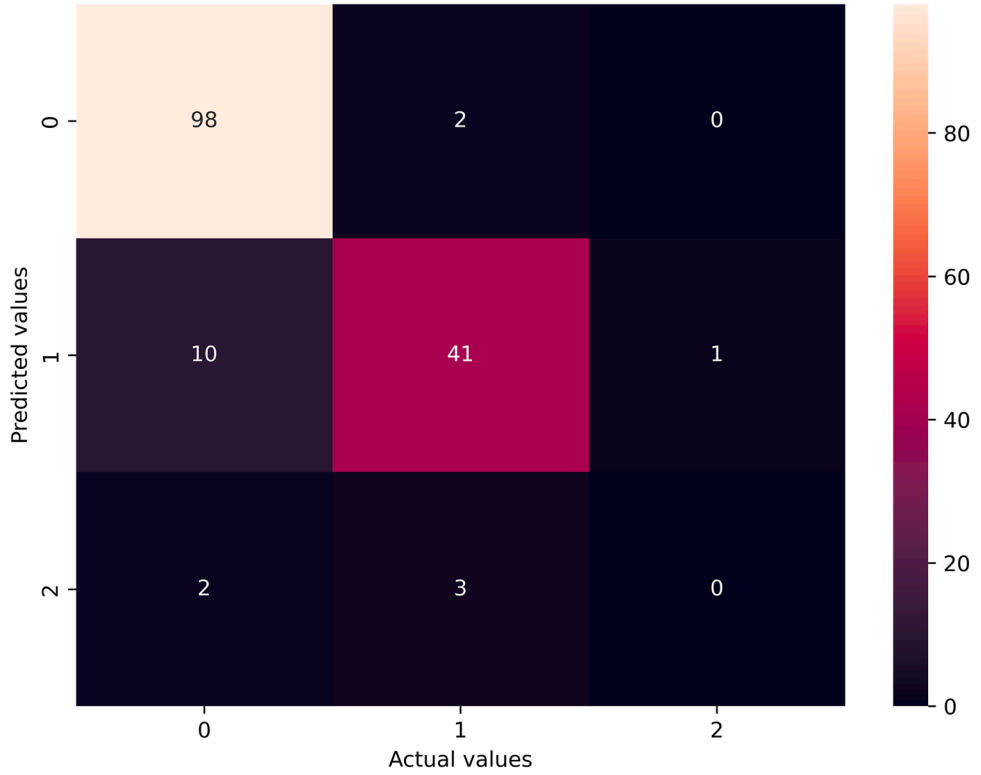


Fig. 13. Confusion matrix by the ensemble stack to predict appendicitis management.

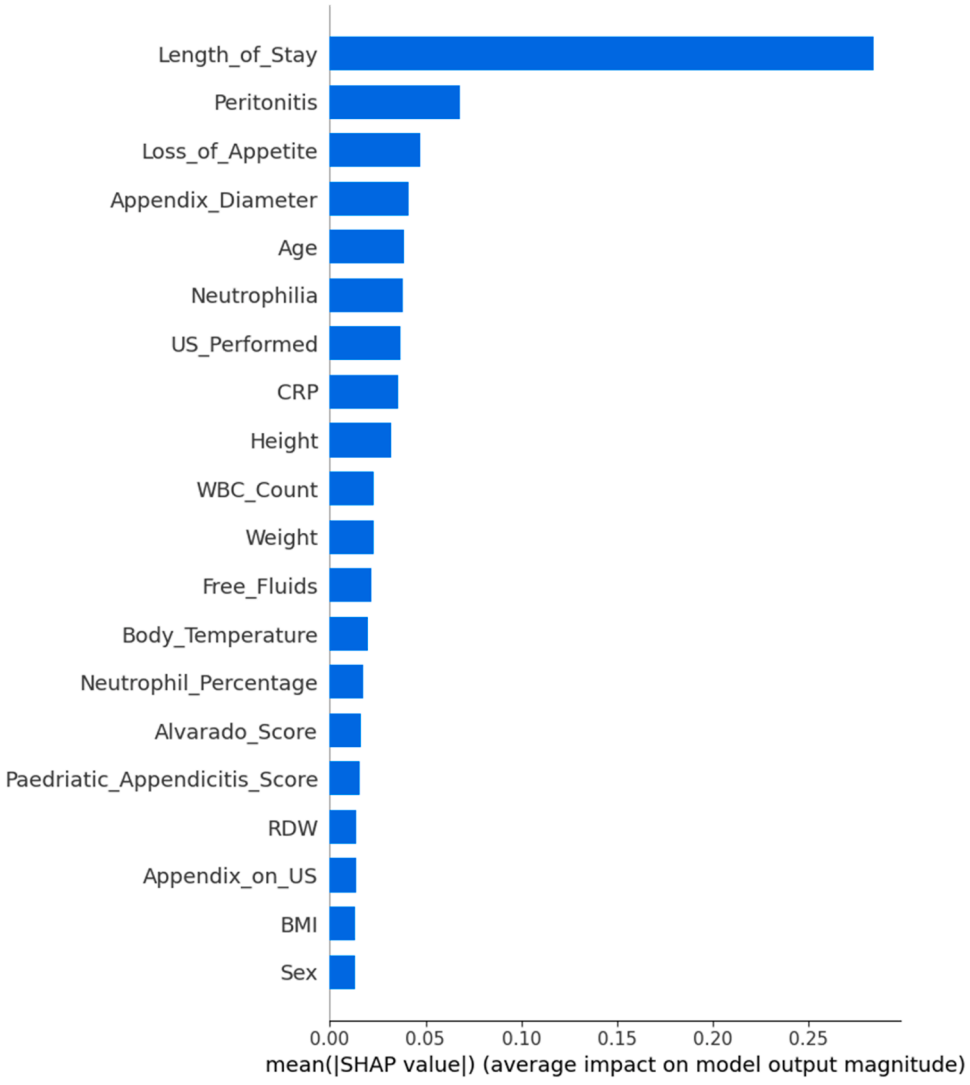


Fig. 14. Feature importance using SHAP bar plot to identify important parameters for appendicitis management.

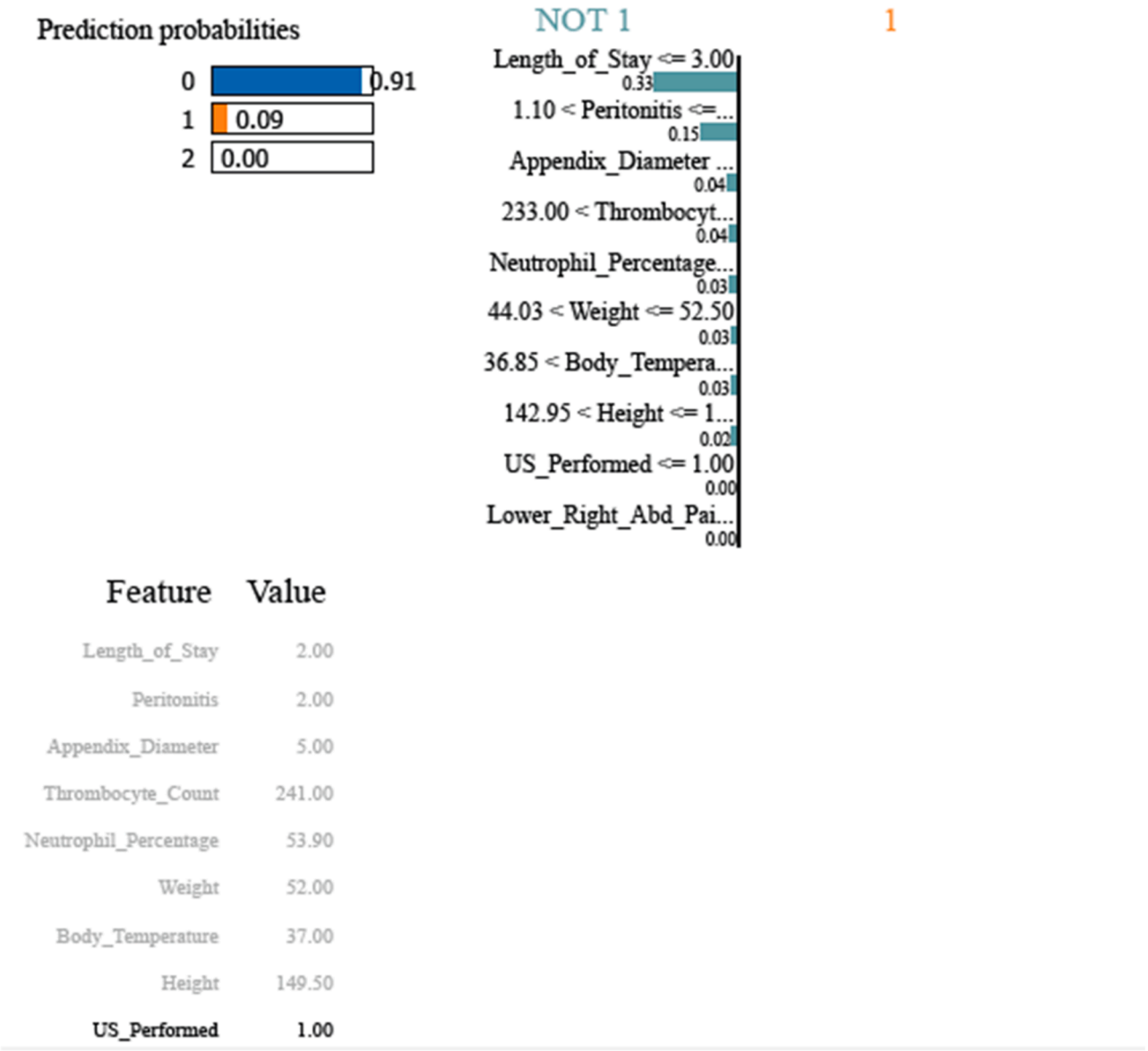


Fig. 15. LIME explanations to identify important parameters for appendicitis management (Patient management prediction-conservative).

y=1.0 (probability 1.000) top features			y=2.0 (probability 0.000) top features			y=3.0 (probability 0.000) top features		
Contribution?	Feature	Value	Contribution?	Feature	Value	Contribution?	Feature	Value
+0.624	Length_of_Stay	3.000	+0.333	<BIAS>	1.000	+0.333	<BIAS>	1.000
+0.333	<BIAS>	1.000	-0.003	US_Performed	1.000	+0.001	Peritonitis	2.000
+0.023	Peritonitis	2.000	-0.003	Thrombocyte_Count	377.000	+0.000	Age	12.194
+0.006	Thrombocyte_Count	377.000	-0.006	Age	12.194	-0.003	US_Performed	1.000
+0.006	Age	12.194	-0.024	Peritonitis	2.000	-0.003	Thrombocyte_Count	377.000
+0.006	US_Performed	1.000	-0.297	Length_of_Stay	3.000	-0.328	Length_of_Stay	3.000

Fig. 16. Eli5 explanations to identify important parameters for appendicitis management.

Table 6
Important attributes in prediction of severity and management of appendicitis by feature analysis and XAI techniques.

Appendicitis severity prediction						
Mutual Information	**Pearson's correlation**	**SHAP**	**LIME**	**Eli5**	**QLattice**	**Anchor**
Length of stay", "Body temperature", "Alvarado score", "Pediatric appendicitis score" and "CRP"	"Length of stay", "CRP", "WBC count", "Peritonitis" and "Alvarado score".	"Length of stay", "Loss of appetite", "CRP", "Peritonitis, "Body temperature", "Alvarado score" and "Pediatric appendicitis score".	"Loss of appetite", "Length of stay", "Peritonitis", "Nausea", "Pediatric appendicitis score" and "Body temperature"	"Length of stay", "Age", "CRP" and "Red blood cell count"	"Length of stay "and "White blood cell count"	"CRP", "Loss of appetite", "Length of stay" and "Nausea"
Appendicitis management prediction						
Mutual Information	**Pearson's correlation**	**SHAP**	**LIME**	**Eli5**	**QLattice**	**Anchor**
"Length of stay", "body temperature", "C-reactive protein (CRP)", "Height" and "Red blood cell distribution width (RDW) with "Length of stay"	"Length of stay"," peritonitis", "appendix diameter", "Alvarado score" and "CRP"	"Length of stay", "Peritonitis", "Loss of appetite", "Appendix diameter" and "Age"	"Length of stay", "Peritonitis", "Appendix diameter", "Thrombocyte count" and "Neutrophil percentage"	"Length of stay", "Peritonitis", "Thrombocyte count" and "Age"	The module did not support multi classification	The module did not support multi classification

Table 8
Comparison of results.

Reference	**Dataset size**	**ML models used**	**Results obtained**	**XAI techniques used**
[20]	332 patients	DT model	AUC – 0.91 Sensitivity- 91.8 %	Feature importance using DT
[22]	551 patients	Three classifiers	Sensitivity – 97 %	Extensive analysis using SHAP
[42]	430 patients	Random forest	Positive predictive value for management– 91 % Positive predictive value for severity– 94 %	Feature importance using random forest
Our study	782 patients	Several classifiers including a stacked ensemble	Severity prediction accuracy - 94 % Management prediction accuracy – 89 %	SHAP, LIME, Eli5, QLattice and Anchor

Ethical approval

We have obtained the data from UCI Machine learning repository. The datasets can be used without ethical clearance. Hence, no ethical clearance was required. We have mentioned the same in the paper.

Funding

There are no funders to report for this submission.

CRediT authorship contribution statement

Pavanya Manohar: Writing – original draft, Software. **Anisha Palkar:** Visualization, Resources, Investigation. **Niranjana Sampathila:** Supervision, Project administration. **Venkatesh Bhandage:** Validation, Formal analysis. **Krishnaraj Chadaga:** Writing – review & editing, Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

We would like to express our gratitude to Manipal Academy of Higher Education for providing us with a platform to conduct this study.

Data availability

Data will be made available on request.

References

[1] J. De Jonge, J.C. Scheijmans, C.C. Van Rossem, A.A. Van Geloven, M. A. Boermeester, W.A. Bemelman, Normal inflammatory markers and acute appendicitis: a national multicentre prospective cohort analysis, Int. J. Colorectal Dis. 36 (7) (2021 Jul) 1507–1513, https://doi.org/10.1007/s00384-021-03933-7.
[2] J.J. Kim, B.H. Dobson, L.H. Ng C, D.W. Thong, T. Arthur, D. Parker, Q.U.E.S. T. Collaboration, T. Anwari, L. Archer, M. Auld, D. Bagguley, Can normal inflammatory markers rule out acute appendicitis? The reliability of biochemical investigations in diagnosis, ANZ J. Surg. 90 (10) (2020 Oct) 1970–1974, https://doi.org/10.1111/ans.15559.
[3] Z. Pogorelić, V. Ercegović, M. Bašković, M. Jukić, I. Karaman, I. Mrklić, Incidence and management of appendiceal neuroendocrine tumors in pediatric population: a bicentric experience with 6285 appendectomies, Children 10 (12) (2023 Dec 8) 1899.
[4] Z. Pogorelić, T. Čohadžić, A bizarre cause of acute appendicitis in a pediatric patient: an ingested tooth, Children 10 (1) (2023 Jan 4) 108.
[5] Z. Pogorelić, V. Babić, M. Bašković, V. Ercegović, I. Mrklić, Management and incidence of Enterobius vermicularis infestation in appendectomy specimens: a cross-sectional study of 6359 appendectomies, J. Clin. Med. 13 (11) (2024 May 29) 3198.
[6] D.T. Gray, T. Mizrahi, Trends in appendicitis and perforated appendicitis prevalence in children in the United States, 2001-2015, JAMA Network Open 3 (10) (2020 Oct 1) e2023484, https://doi.org/10.1001/jamanetworkopen.2020.23484. -.
[7] Z. Pogorelić, J. Mihanović, S. Ninčević, B. Lukšić, S. Elezović Baloević, O. Polašek, Validity of appendicitis inflammatory response score in distinguishing perforated from non-perforated appendicitis in children, Children 8 (4) (2021 Apr 19) 309.
[8] M. Jukić, A. Tesch, J. Todorić, T. Šušnjar, K.P. Milunović, T. Barić, Z. Pogorelić, Same-day discharge after laparoscopic appendectomy for simple appendicitis in pediatric patients—Is it possible? Children 9 (8) (2022 Aug 12) 1220.
[9] J. Kleif, L.C. Thygesen, I. Gögenur, Moving from an era of open appendectomy to an era of laparoscopic appendectomy: a nationwide cohort study of adult patients undergoing surgery for appendicitis, Scandinavian J. Surg. 110 (4) (2021 Dec) 512–519, https://doi.org/10.1177/1457496921992615.
[10] Z. Küçükakçali, S. Akbulut, C. Çolak, Value of fecal calprotectin in prediction of acute appendicitis based on a proposed model of machine learning, Turkish J. Trauma Emerg. Surg. 29 (6) (2023 Jun) 655.
[11] M.M. Mijwil, K. Aggarwal, A diagnostic testing for people with appendicitis using machine learning techniques, Multimed. Tools Appl. 81 (5) (2022 Feb) 7011–7023, https://doi.org/10.1007/s11042-022-11939-8.
[12] H. Omotunde, M.R. Mouhamed, The modern impact of artificial intelligence systems in healthcare: a concise analysis, Mesop. J. Artif. Intell. Healthcare 2023 (2023 Nov 22) 66–70, https://doi.org/10.58496/MJAIH/2023/013.

[13] S. Akbulut, F.H. Yagin, I.B. Cicek, C. Koc, C. Colak, S. Yilmaz, Prediction of perforated and nonperforated acute appendicitis using machine learning-based explainable artificial intelligence, Diagnostics 13 (6) (2023 Mar 19) 1173, https://doi.org/10.3390/diagnostics13061173.
[14] S. Akbulut, C. Koç, T.T. Şahin, E. Şahin, A. Tuncer, K. Demyati, E.T. Şamdancı, C. Çolak, S. Yılmaz, An investigation into the factors predicting acute appendicitis and perforated appendicitis, Turkish J. Trauma Emerg. Surg./Ulusal Travma ve Acil Cerrahi Dergisi 27 (4) (2021 Jul 1).
[15] S. Akbulut, Z.Ş. Bahçe, T. Öztaş, S. Gümüş, N. Söğütçü, H. Sakarya, A.F. Gök, Y. Yağmur, Assessment of demographic, clinical and histopathological features of patients who underwent appendectomy due to a presumed diagnosis of acute appendicitis, Turkish J. Trauma Emerg. Surg./Ulusal Travma ve Acil Cerrahi Dergisi 27 (3) (2021 May 1).
[16] A. Emre, S. Akbulut, Z. Bozdag, M. Yilmaz, M. Kanlioz, R. Emre, N. Sahin, Routine histopathologic examination of appendectomy specimens: retrospective analysis of 1255 patients, Int. Surg. 98 (4) (2013 Oct 1) 354–362.
[17] M. Yilmaz, S. Akbulut, K. Kutluturk, N. Sahin, E. Arabaci, C. Ara, S. Yilmaz, Unusual histopathological findings in appendectomy specimens from patients with suspected acute appendicitis, World J. Gastroenterol 19 (25) (2013) 4015–4022.
[18] R. Marcinkevičs, P.R. Wolfertstetter, U. Klimiene, K. Chin-Cheong, A. Paschke, J. Zerres, M. Denzinger, D. Niederberger, S. Wellmann, E. Ozkan, C. Knorr, Interpretable and intervenable ultrasonography-based machine learning models for pediatric appendicitis, Med. Image Anal. 91 (2024 Jan 1) 103042, https://doi.org/10.1016/j.media.2023.103042.
[19] W. Abu-Ashour, S. Emil, D. Poenaru, Using artificial intelligence to label free-text operative and ultrasound reports for grading pediatric appendicitis, J. Pediatr. Surg. (2024 Feb 2), https://doi.org/10.1016/j.jpedsurg.2024.01.033.
[20] J. Byun, S. Park, S.M. Hwang, Diagnostic algorithm based on machine learning to predict complicated appendicitis in children using CT, laboratory, and clinical features, Diagnostics 13 (5) (2023 Mar 1) 923, https://doi.org/10.3390/diagnostics13050923.
[21] L. Shahmoradi, R. Safdari, M.M. Mirhosseini, S. Rezayi, M. Javaherzadeh, Development and evaluation of a clinical decision support system for early diagnosis of acute appendicitis, Sci. Rep. 13 (1) (2023 Nov 11) 19703, https://doi.org/10.1038/s41598-023-46721-9.
[22] I. Males, Z. Boban, M. Kumric, et al., Applying an explainable machine learning model might reduce the number of negative appendectomies in pediatric patients with a high probability of acute appendicitis, Sci. Rep. 14 (2024) 12772.
[23] R. Marcinkevičs, P. Reis Wolfertstetter, U. Klimiene, E. Ozkan, K. Chin-Cheong, A. Paschke, J. Zerres, M. Denzinger, D. Niederberger, S. Wellmann, C. Knorr, J. E. Vogt, Regensburg pediatric Appendicitis dataset (1.01) [Data set], Zenodo (2023), https://doi.org/10.5281/zenodo.7669442.
[24] A. Jadhav, D. Pramod, K. Ramanathan, Comparison of performance of data imputation methods for numeric dataset, Appl Artif. Intell. 33 (10) (2019 Aug 24) 913–933, https://doi.org/10.1080/08839514.2019.1637138.
[25] M.M. Ahsan, M.P. Mahmud, P.K. Saha, K.D. Gupta, Z. Siddique, Effect of data scaling methods on machine learning algorithms and model performance, Technologies 9 (3) (2021 Jul 24) 52, https://doi.org/10.3390/technologies9030052.
[26] T. Al-Shehari, R.A. Alsowail, An insider data leakage detection using one-hot encoding, synthetic minority oversampling and machine learning techniques, Entropy 23 (10) (2021 Sep 27) 1258, https://doi.org/10.3390/e23101258.
[27] Y. Chen, R. Chang, J. Guo, Effects of data augmentation method borderline-SMOTE on emotion recognition of EEG signals based on convolutional neural network, IEEE Access 9 (2021 Mar 24) 47491–47502, https://doi.org/10.1109/ACCESS.2021.3068316.
[28] M.Y. Shams, A.M. Elshewey, E.S. El-kenawy, A. Ibrahim, F.M. Talaat, Z. Tarek, Water quality prediction using machine learning models based on grid search method, Multimed. Tools Appl. 83 (12) (2024 Apr) 35307–35334, https://doi.org/10.1007/s11042-023-16737-4.
[29] M. Sugiyama, Machine learning with squared-loss mutual information, Entropy 15 (1) (2012 Dec 27) 80–112, https://doi.org/10.3390/e15010080.
[30] H. Zhou, Z. Deng, Y. Xia, M. Fu, A new sampling method in particle filter based on Pearson correlation coefficient, Neurocomputing 216 (2016 Dec 5) 208–215, https://doi.org/10.1016/j.neucom.2016.07.036.
[31] M.S. Rahman, H.R. Rahman, J. Prithula, M.E. Chowdhury, M.U. Ahmed, J. Kumar, M. Murugappan, M.S. Khan, Heart failure emergency readmission prediction using stacking machine learning model, Diagnostics 13 (11) (2023 Jun 2) 1948, https://doi.org/10.3390/diagnostics13111948.
[32] V. Vivek Khanna, K. Chadaga, N. Sampathila, S. Prabhu, P.R. Chadaga, D. Bhat, S. KS, Explainable artificial intelligence-driven gestational diabetes mellitus prediction using clinical and laboratory markers, Cogent Eng. 11 (1) (2024 Dec 31) 2330266, https://doi.org/10.1080/23311916.2024.2330266.
[33] K. Chadaga, S. Prabhu, N. Sampathila, R. Chadaga, S. Umakanth, D. Bhat, G.S. SK, Explainable artificial intelligence approaches for COVID-19 prognosis prediction using clinical markers, Sci. Rep 14 (1) (2024 Jan 20) 1783, https://doi.org/10.1038/s41598-024-52428-2.
[34] T.C. Chen, H.C. Wu, M.C. Chiu, A deep neural network with modified random forest incremental interpretatioproach for diagnosing diabetes in smart healthcare, Appl. Soft Comput. 152 (2024 Feb 1) 111183, https://doi.org/10.1016/j.asoc.2023.111183.
[35] S. Prusty, S. Patnaik, S.K. Dash, S.G.P. Prusty, J. Rautaray, G. Sahoo, Predicting cervical cancer risk probabilities using advanced H20 AutoML and local interpretable model-agnostic explanation techniques, Peer. J. Comput. Sci. 10 (2024) e1916, https://doi.org/10.7717/peerj-cs.1916.
[36] B.S. Darshan, N. Sampathila, M.G. Bairy, S. Belurkar, S. Prabhu, K. Chadaga, Detection of anemic condition in patients from clinical markers and explainable artificial intelligence, Technol. Health Care (2024 Feb 1) 1–4, https://doi.org/10.3233/THC-231207 (Preprint).
[37] A. Palkar, C.C. Dias, K. Chadaga, N. Sampathila, Empowering glioma prognosis with transparent machine learning and interpretative insights using explainable AI, IEEE Access 12 (2024 Feb 26) 31697–31718. Palkar A, Dias CC, Chadaga K, Sampathila N. Empowering Glioma Prognosis With Transparent Machine Learning and Interpretative Insights Using Explainable AI. IEEE Access. 2024 Feb 26;12:31697-718.
[38] T. Spinner, U. Schlegel, H. Schäfer, El-Assady M. explAIner, A visual analytics framework for interactive and explainable machine learning, IEEE transactions on visualization and computer graphics 26 (1) (2019 Aug 20) 1064–1074, https://doi.org/10.1109/TVCG.2019.2934629.
[39] ĄA Kaminskas, R. Lukšaitė-Lukštė, E. Jasiūnas, A. Samuilis, V. Augustinavičius, M. Kryžauskas, K. Strupas, T. Poškus, The dynamics of inflammatory markers in patients with suspected acute appendicitis, Medicina 57 (12) (2021 Dec 20) 1384, https://doi.org/10.3390/medicina57121384.
[40] A. Güngör, A. Göktuğ, M.M. Güneylioğlu, R.M. Yaradılmış, I. Bodur, B. Öztürk, İ. Karaman, C.D. Karacan, N. Tuygun, Utility of biomarkers in predicting complicated appendicitis: can immature granulocyte percentage and C-reactive protein be used? Postgrad. Med. 133 (7) (2021 Oct 3) 817–821, https://doi.org/10.1080/00325481.2021.1948306.
[41] I. Maleš, M. Kumrić, A. Huić Maleš, I. Cvitković, R. Šantić, Z. Pogorelić, J. Božić, A systematic integration of artificial intelligence models in appendicitis management: a comprehensive review, Diagnostics 15 (7) (2025 Mar 28) 866.
[42] R. Marcinkevics, P. Reis Wolfertstetter, S. Wellmann, C. Knorr, J.E Vogt, Using machine learning to predict the diagnosis, management and severity of pediatric appendicitis, Front. Pediatr. 9 (2021 Apr 29) 662183, https://doi.org/10.3389/fped.2021.662183.